

Reason about arithmetic relationships

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: If $4,725 / 35 = 135$, find $4,725 / 350$. Copy or complete the first step.

2. If $48 \times 125 = 6,000$, find 4.8×125 .

3. Which is greater: $5/6$ of 420 or 0.82 of 420?

4. Miro says: Miro changes one factor by ten but expects the product to stay unchanged. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: If $4,725 / 35 = 135$, find $4,725 / 350$. Copy or complete the first step.

Answer 13.5.

2. If $48 \times 125 = 6,000$, find 4.8×125 .

Answer 600.

3. Which is greater: $5/6$ of 420 or 0.82 of 420?

Answer $5/6$ of 420 = 350; 0.82 of 420 = 344.4, so $5/6$ is greater.

4. Miro says: Miro changes one factor by ten but expects the product to stay unchanged. Is this correct? Explain.

Answer Changing a factor by a power of ten changes the product by the same scale unless another factor compensates.

Reason about arithmetic relationships

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. If $4,725 \div 35 = 135$, find $4,725 \div 350$.

6. Check this result using an inverse, estimate or known fact: Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

2. If $48 \times 125 = 6,000$, find 4.8×125 .

7. Create a chain of three related facts from $2,304 \times 27 = 62,208$.

3. Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

8. Identify and correct this misconception: Miro changes one factor by ten but expects the product to stay unchanged.

4. Solve this independently and show every stage: If $4,725 \div 35 = 135$, find $4,725 \div 350$.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: If $48 \times 125 = 6,000$, find 4.8×125 .

10. Write and solve a similar question to: Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

Teacher answers and checking prompts

1. If $4,725 / 35 = 135$, find $4,725 / 350$.
Answer 13.5.
2. If $48 \times 125 = 6,000$, find 4.8×125 .
Answer 600.
3. Which is greater: $5/6$ of 420 or 0.82 of 420?
Answer $5/6$ of 420 = 350; 0.82 of 420 = 344.4, so $5/6$ is greater.
4. Solve this independently and show every stage: If $4,725 / 35 = 135$, find $4,725 / 350$.
Answer 13.5.
5. Use a second method or representation for: If $48 \times 125 = 6,000$, find 4.8×125 .
Answer Expected result: 600. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Which is greater: $5/6$ of 420 or 0.82 of 420?
Answer Expected result: $5/6$ of 420 = 350; 0.82 of 420 = 344.4, so $5/6$ is greater. A valid check must be shown.
7. Create a chain of three related facts from $2,304 \times 27 = 62,208$.
Answer Answers vary, for example $62,208 / 27 = 2,304$ and $230.4 \times 27 = 6,220.8$.
8. Identify and correct this misconception: Miro changes one factor by ten but expects the product to stay unchanged.
Answer Changing a factor by a power of ten changes the product by the same scale unless another factor compensates.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Which is greater: $5/6$ of 420 or 0.82 of 420?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Reason about arithmetic relationships

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Create a chain of three related facts from $2,304 \times 27 = 62,208$.

6. Create a new example that tests the same idea as: Which is greater: $\frac{5}{6}$ of 420 or 0.82 of 420?

2. Prove the answer to this question, rather than only calculating it: If $4,725 \div 35 = 135$, find $4,725 \div 350$.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: If $48 \times 125 = 6,000$, find 4.8×125 .

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: $\frac{5}{6}$ of 420 = 350; 0.82 of 420 = 344.4, so $\frac{5}{6}$ is greater.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro changes one factor by ten but expects the product to stay unchanged.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Create a chain of three related facts from $2,304 \times 27 = 62,208$.
Answer Answers vary, for example $62,208 / 27 = 2,304$ and $230.4 \times 27 = 6,220.8$.
2. Prove the answer to this question, rather than only calculating it: If $4,725 / 35 = 135$, find $4,725 / 350$.
Answer Expected result: 13.5. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: If $48 \times 125 = 6,000$, find 4.8×125 .
Answer Expected result: 600. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: $5/6$ of 420 = 350; 0.82 of 420 = 344.4, so $5/6$ is greater.
Answer One valid reconstruction is: Which is greater: $5/6$ of 420 or 0.82 of 420? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro changes one factor by ten but expects the product to stay unchanged.
Answer Changing a factor by a power of ten changes the product by the same scale unless another factor compensates.
6. Create a new example that tests the same idea as: Which is greater: $5/6$ of 420 or 0.82 of 420?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.